

**General Site Information:
Meteorological, Geological
And Utilities Data**

Project No.: 01000
KPI Doc No.: 01000-EM-SPC-003
Contr. Doc No.:
Rev.: R00
Page: 2 / 21

CONTENTS

1	SITE METEOROLOGICAL DATA.....	3
1.1	Climatic Data for The Warm Season Design	3
1.2	Climatic Data for The Cold Season.....	3
1.3	Average Monthly and Annual Air Temperature	4
1.4	Wet Bulb Temperature	4
1.5	Rainfall	4
1.6	Snowfall, Snow Cover and Glaze Phenomena	4
1.7	Wind Pressure, Wind Velocity, Wind Direction and wind load	5
1.8	Atmosphere.....	6
1.9	Statistical Climatic Data From Atyrau Weather Station Atyrau	6
2	SITE GEOLOGICAL AND HYDROLOGICAL DATA.....	14
2.1	Site Locations IPC	14
2.2	Existing Sites	14
2.3	Geotechnical Investigations At IPC Site IPC	14
2.4	Hydrological Studies	15
2.5	Earthquake.....	15
3	SITE UTILITY SYSTEMS DATA.....	15
3.1	Utilities: General.....	15
3.2	Utilities: Water	16
3.3	Utilities: Air	19
3.4	Utilities: Fuels.....	20
3.5	Utilities: Electricity	21

**General Site Information:
Meteorological, Geological
And Utilities Data**

Project No.:	01000
KPI Doc No.:	01000-EM-SPC-003
Contr. Doc No.:	
Rev.:	R00
Page:	3 / 21

1 SITE METEOROLOGICAL DATA

The following data is deemed to be applicable to all the Facilities whether situated at Karabatan, Atyrau region, West Kazakhstan.

The data are coming from the following various sources: SNiP, Atyrau weather station, data collected for the site investigation report.

The region is a low flat country or a slightly elevated flat country located in a semidesert and desert zone.

1.1 Climatic Data for The Warm Season Design

Abstracted from SNiP 2.04-01-2001)

Max. temperature (°C) with probability of 0.95	31.2
Max. temperature (°C) with probability of 0.98	34.2
Average max. temperature (°C) in the warmest month	32.1
Absolute max. temperature (°C)	43.0
Average difference between max. and min. temperature (°C) over 24 hrs for the warmest month	13.1
Average monthly relative humidity for warmest month at 15:00 hrs	29%
Barometric Pressure (mBar)	1020

1.2 Climatic Data for The Cold Season

(Abstracted from SNiP 2.04-01-2001)

Min. temperature (°C) of the coldest day with probability of 0.98	-33
Min. temperature (°C) of the coldest day with probability of 0.92	-30
Min. temperature (°C) of the coldest 5 days with probability of 0.98	-28
Min. temperature (°C) of the coldest 5 days with probability of 0.92	-25
Temperature (°C) above which is reached 94% of the time	-8
Absolute min. temperature (°C)	-38
Average difference between max. and min. temperature (°C) over 24 hrs for the coldest month	8.1

**General Site Information:
Meteorological, Geological
And Utilities Data**

Project No.: 01000
KPI Doc No.: 01000-EM-SPC-003
Contr. Doc No.:
Rev.: R00
Page: 4 / 21

Number of days when min temperature is $\leq 0^{\circ}\text{C}$	125
Average temperature ($^{\circ}\text{C}$) on the days when min. temperature is $\leq 0^{\circ}\text{C}$	-6.4
Number of days when min temperature is $\leq 8^{\circ}\text{C}$	177
Average temperature ($^{\circ}\text{C}$) on the days when min. temperature is $\leq 8^{\circ}\text{C}$	-3.4
Number of days when min temperature is $\leq 10^{\circ}\text{C}$	191
Average temperature ($^{\circ}\text{C}$) on the days when min. temperature is $\leq 10^{\circ}\text{C}$	-2.4
Average monthly relative humidity for coldest month	84%
Average monthly relative humidity for coldest month at 15:00 Hrs	75%

1.3 Average Monthly and Annual Air Temperature

(abstracted from SNiP 2.04-01-2001)

Average Monthly and Annual Air Temperatures ($^{\circ}\text{C}$)

Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	Average annual
- 9.6	-8.7	-1.5	9.6	18.2	23.4	25.7	23.7	16.8	8.2	0.4	-5.6	8.4

1.4 Wet Bulb Temperature

For Design purpose, Wet Bulb Temperature shall be considered as 29°C .

Wet bulb average month air temperature are given in section 1.9.2.

For Design of cooling tower, Wet Bulb Temperature shall be 27°C .

1.5 Rainfall

Average Annual Precipitation Quantity (mm)

Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	Year	Cold Period	Warm Period
14	12	13	14	16	19	16	12	12	13	13	18	172	70	102

Daily max: 58 mm

1.6 Snowfall, Snow Cover and Glaze Phenomena

Snow fall: 50 – 200 mm per year

Snow load: 1.32 kN/m^2

**General Site Information:
Meteorological, Geological
And Utilities Data**

Project No.: 01000
KPI Doc No.: 01000-EM-SPC-003
Contr. Doc No.:
Rev.: R00
Page: 5 / 21

Weight of Snow Cover Area: Region I (abstracted from SNIP 2.01.07-85)

1.6.1 Snow Cover

Mean date of formation and destruction of steady snow cover	Average height for winter, cm	Maximum height for winter, cm
10th December – 4th March	10	33

1.6.2 Glaze Phenomena

Region through glaze wall thickness	Standard glaze wall thickness with recurrence one in 5 years, mm	Standard glaze wall thickness with recurrence once in 10 years, mm
II	5	10

1.7 Wind Pressure, Wind Velocity, Wind Direction and wind load

Wind Area – abstracted from Snip 2.01.07-85:

Average wind speed during winter period: Region 6

Wind pressure: Region III

1.7.1 Monthly Mean and Annual Wind Velocity, m/s

Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	Year
4.6	5.2	5.2	5.1	5.2	4.6	4.3	3.9	3.7	4.3	4.5	4.8	4.6

1.7.2 Wind Pressure

Wind area	Wind Pressure q0, pressure N/m ² (wind velocity V, m/s) with recurrence		
III	Once in 5 years 450(27)	Once in 10 years 516(29)	Once in 15 years 552(30)

1.7.3 Dominant wind direction for Warm and cold season

Dominant wind direction for Warm season June – August South-West, North-West

Dominant wind direction for cold season December – February East

**General Site Information:
Meteorological, Geological
And Utilities Data**

Project No.: 01000
KPI Doc No.: 01000-EM-SPC-003
Contr. Doc No.:
Rev.: R00
Page: 6 / 21

1.7.4 Wind load

The site basic wind speed “Vb” is 29.4m/sec with a return period of 50 years.

1.8 Atmosphere

1 Extreme moisture (tropical climate): none

2 Marine exposure (salt spray): Yes

3 Sand storms: Frequency: 43 days/year; Average solids loading in air: 26 mg/m³; Max. solids loading in air: 240 mg/m³ at 3m, 50 mg/m³ at 6m

4 Ice storms occur periodically

5 Copper-attacking fumes (ammonia, sulphur, etc.): SO₂: [will be provided by TTC] (maximum)

6 Exposure to conductive or corrosive dusts (eg. carbon, iron oxide ammonium nitrates or phosphates, etc.): NO_x: [will be provided by TTC] gm (maximum) at [will be provided by TTC]

7 Exposure to corrosive agents (nitric or sulphuric acids, chlorine, caustic, etc.): SO₂ and NO_x as above

8 Exposure to other pollutants originating from surrounding industrial plant: [will be provided by TTC]

1.9 Statistical Climatic Data From Atyrau Weather Station Atyrau

1.9.1 Air Temperature °C (averaging period years 1986-2007)

Average month and annual air temperature (°C)

Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	Year
-9.0	-8.4	-1.4	10.0	18.3	23.4	25.8	23.8	17.0	8.3	0.5	-5.5	8.6

Average maximum air temperature (°C)

Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	Year
-4.9	-3.9	3.7	16.4	24.6	29.8	32.3	30.6	23.7	14.0	4.8	-1.8	14.1

**General Site Information:
Meteorological, Geological
And Utilities Data**

Project No.: 01000
KPI Doc No.: 01000-EM-SPC-003
Contr. Doc No.:
Rev.: R00
Page: 7 / 21

Average minimum air temperature (°C)

Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	Year
-12.7	-12.5	-5.7	4.3	12.0	17.0	19.3	17.2	10.8	3.3	-3.3	-8.7	3.4

Average from absolute maximum air temperature (°C)

Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	Year
3	4	14	26	32	37	39	37	32	22	13	5	39

Note: Occasionally absolute ambient air temperature of 50°C is occurring. However, it is not advisable to design the plant at 50°C, and design ambient air temperature process plants and utilities shall remain unchanged at 43 °C (Absolute max. temperature, see item 1.1). As an operational issue, all air utilization systems, i.e. air compressors should be able to sustain occasional high ambient air temperatures of 50°C.

Average from absolute minimum air temperature (°C)

Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	Year
-25	-24	-17	-4	4	10	14	10	3	-6	-14	-21	-28

1.9.2 Wet Bulb Average Month Air Temperature (°C) for The period 1986-2007

Year	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
1998	-4.3	-3.9	-2.0	6.1	11.5	18.6	19.5	17.1	10.3	6.4	0.0	-2.6
1999	-3.5	-1.4	-1.0	7.4	10.2	16.1	18.2	18.7	11.4	8.6	-1.1	-1.0
2000	-3.4	-1.5	1.3	10.2	10.3	16.5	19.1	17.6	11.6	4.6	-1.4	-2.3
2001	-2.7	-3.2	1.9	8.7	13.8	16.0	18.3	15.8	12.2	5.6	2.7	-5.1
2002	-1.9	0.2	3.5	6.0	10.3	15.3	18.5	15.3	13.7	7.0	1.8	-6.9
2003	-3.5	-5.3	-2.4	5.2	13.3	14.7	18.5	18.8	12.6	7.8	1.8	-2.8
2004	-3.9	-1.5	2.6	7.2	12.8	16.2	18.7	17.4	13.4	7.0	2.6	-2.5
2005	-3.3	-4.7	-0.7	7.9	14.9	17.1	19.0	16.4	14.3	8.6	1.2	-0.9
2006	-5.7	-4.0	1.1	8.9	12.8	18.4	16.5	18.2	12.6	8.3	2.1	-1.6
2007	-0.7	-1.7	0.1	7.7	13.6	16.1	18.8	18.1	13.2	6.9	-0.8	-3.6

1.9.3 Atmospheric Pressure On The Sea-level, hPa (Averaging Period Years 1986-2007)

Average month and annual atmospheric pressure on the sea-level

Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	Year
1023.5	1023.7	1022.2	1016.8	1015.3	1010.8	1009.6	1012.3	1017.0	1021.3	1023.7	1023.0	1018.3

Maximum atmospheric pressure on the sea-level

Month	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
	1048.1	1043.6	1041.8	1041.3	1031.9	1023.9	1020.6	1023.8	1034.1	1038.7	1044.8	1049.6
Year	2006	2005	2004	2003	2006	2004	2003	2002	2003	2001	1997	2007

Minimum atmospheric pressure on the sea-level

**General Site Information:
Meteorological, Geological
And Utilities Data**

Project No.: 01000
KPI Doc No.: 01000-EM-SPC-003
Contr. Doc No.:
Rev.: R00
Page: 8 / 21

Month	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
	993.0	989.3	994.9	995.3	991.3	992.2	992.0	997.0	1001.6	994.7	991.5	1000.2
Year	2000	1998	1998	1997	2004	2000	1998	2002	2001	2005	2004	2007

1.9.4 Air Relative Humidity% for the period 1998 – 2007

Average month relative humidity

Year	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
1998	80	77	75	60	42	40	45	44	38	62	77	81
1999	81	80	55	47	49	43	43	43	53	66	82	87
2000	78	75	64	55	56	50	48	45	56	67	72	89
2001	85	80	64	58	55	48	44	42	53	69	77	73
2002	84	85	68	65	58	47	35	40	48	64	77	81
2003	80	70	68	53	54	54	48	48	50	58	86	83
2004	85	77	73	60	47	42	54	39	52	67	80	87
2005	79	70	73	55	45	45	45	42	44	64	76	86
2006	82	88	72	62	53	40	39	35	47	69	84	91
2007	88	83	68	66	50	40	45	33	48	50	83	86

Maximum air relative humidity

Year	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
1998	100	100	96	98	92	90	89	92	88	97	96	100
1999	100	100	100	98	99	94	95	97	95	100	100	98
2000	96	98	97	95	98	98	91	90	93	98	98	99
2001	98	100	98	97	100	95	93	94	97	96	98	96
2002	97	98	96	95	98	97	79	92	96	100	100	100
2003	99	91	94	94	97	94	95	96	91	98	100	100
2004	100	98	98	100	94	94	93	85	94	100	100	100
2005	98	98	98	100	94	92	85	95	90	98	98	100
2006	98	100	100	100	94	90	91	94	94	100	98	100
2007	100	100	98	97	96	84	89	74	97	89	100	100

1.10 Miscellaneous Site Data

1.10.1 Thunderstorm Frequency for multiple years monitoring in Atyrau:

Average number of days with thunderstorms

	MONTH												
Atyrau	I	II	III	IV	V	VI	VII	VIII	IX	X	XI	XII	year
Number of days	-	-	00.3	0.2	1.2	2.6	2.5	1.6	0.6	0.2	0.03	-	9.0

Maximum number of days with thunderstorm

	MONTH												
Atyrau	I	II	III	IV	V	VI	VII	VIII	IX	X	XI	XII	year
Number	-	-	1	1	4	7	8	6	4	2	1	-	19

**General Site Information:
Meteorological, Geological
And Utilities Data**

Project No.: 01000
KPI Doc No.: 01000-EM-SPC-003
Contr. Doc No.:
Rev.: R00
Page: 9 / 21

of days													
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Average duration of thunderstorms in hours

	MONTH												
Atyrau	I	II	III	IV	V	VI	VII	VIII	IX	X	XI	XII	year
Number of days	-	-	00.3	0.2	2.5	5.5	4.5	3.0	1.2	0.1	0.1	-	17.2

1.10.2 Temperature Inversion Occurrence:

Temperature inversion in December 2006

Date	2006					
	Near the ground (0 to 1 km)			Atmospheric (1 to 30 km)		
	Inversion	Height (m)	Temperature (°C)	Inversion	Height (m)	Temperature (°C)
1 Dec	yes	0,314	-4,9	yes	10,956	-61,2
2 Dec	no			yes	20,983	-65,2
3 Dec	no			no		
4 Dec	yes	0,303	0,7	yes	22,045	-62,5
5 Dec	yes	0,504	0,8	yes	11,712	-67,4
6 Dec	yes	0,212	3,3	yes	11,877	-67,8
7 Dec	yes	0,454	4,0	yes	11,478	-65,7
8 Dec	yes	0,837	0,2	yes	11,547	-62,4
9 Dec	yes	0,764	-0,6	yes	15,826	-60,8
10 Dec	yes	0,914	1,9	yes	11,145	-60,4
11 Dec	yes	0,902	0,7	yes	11,090	-59,5
12 Dec	no			yes	10,958	-59,2
13 Dec	yes	1,000	2,0	yes	11,240	-61,3
14 Dec	no			yes	11,002	-68,2
15 Dec	yes	0,205	-6,9	yes	10,511	-61,0
16 Dec	no	1		no		
17 Dec	yes	0,458	3,2	yes	11,963	-65,7
18 Dec	yes	0,463	7,1	yes	11,218	-61,3
19 Dec	yes	0,577	0,9	yes	11,368	-65,6
20 Dec	no			yes	11,504	-67,9
21 Dec	no			yes	10,566	-65,3
22 Dec	no			yes	10,455	-61,1
23 Dec	yes	0,500	-5,9	yes	11,385	-63,6
24 Dec	no			yes	8,989	-52,0
25 Dec	no			yes	9,821	-57,0
26 Dec	no			yes	10,758	-63,7
27 Dec	yes	0,469	-8,2	yes	9,485	-61,0
28 Dec	no			yes	10,687	-65,1
29 Dec	no			yes	11,064	-66,1
30 Dec	no			yes	10,573	-64,2
31 Dec	no			yes	8,251	-53,0

**General Site Information:
Meteorological, Geological
And Utilities Data**

Project No.: 01000
KPI Doc No.: 01000-EM-SPC-003
Contr. Doc No.:
Rev.: R00
Page: 10 / 21

Temperature inversion in December 2007

Date	2007					
	Near the ground (0 to 1 km)			Atmospheric (1 to 30 km)		
	Inversion	Height (m)	Temperature (°C)	Inversion	Height (m)	Temperature (°C)
1 Dec	yes	0,302	1,5	yes	11,117	-64,1
2 Dec	yes	0,059	-0,1	yes	17,202	-68,5
3 Dec	yes	0,432	-2,8	yes	19,354	-67,5
4 Dec	yes	0,505	-9,5	yes	11,149	-66,6
5 Dec	yes	0,519	-6,2	yes	17,428	-64,7
6 Dec	yes	0,464	-6,7	yes	17,695	-66,5
7 Dec	yes	0,483	-6,4	yes	11,363	-64,8
8 Dec	yes	0,486	-4,0	yes	11,755	-60,9
9 Dec	no			no		
10 Dec	yes	0,295	-5,0	yes	10,736	-57,1
11 Dec	yes	0,168	-6,3	yes	16,170	-63,5
12 Dec	yes	0,187	-7,3	yes	16,391	-60,9
13 Dec	yes	0,478	0,1	yes	9,469	-57,7
14 Dec	yes	0,317	-6,4	yes	11,045	-60,2
15 Dec	yes	0,243	-5,6	yes	9,931	-61,5
16 Dec	yes	0,308	-5,1	yes	10,704	-67,0
17 Dec	yes	0,644	-6,0	yes	10,020	-59,8
18 Dec	yes	0,255	-7,1	yes	9,586	-61,0
19 Dec	yes	0,621	-7,3	yes	18,814	-70,0
20 Dec	yes	0,819	-6,1	yes	9,013	-57,2
21 Dec	no			no		
22 Dec	yes	0,553	-4,3	no		
23 Dec	no			yes	18,611	-69,5
24 Dec	yes	0,221	-17,1	yes	21,553	-69,3
25 Dec	yes	0,729	-9,0	yes	11,245	-66,9
26 Dec	yes	0,048	-17,0	yes	10,813	-66,1
27 Dec	yes	0,130	-13,4	yes	10,593	-61,6
28 Dec	yes	0,464	-6,8	yes	10,341	-60,0
29 Dec	yes	0,302	-8,2	yes	10,369	-61,5
30 Dec	yes	0,861	-2,4	yes	10,943	-64,2
31 Dec	yes	0,067	-17,6	yes	10,491	-63,6

Temperature inversion in July 2008

Date	2008					
	Near the ground (0 to 1 km)			Atmospheric (1 to 30 km)		
	Inversion	Height (m)	Temperature (°C)	Inversion	Height	Temperature (°C)
1 Jul	yes	0,178	21,7	yes	10,728	-49,9
2 Jul	yes	0,275	18,4	yes	16,191	-54,4
3 Jul	yes	0,353	21,1	yes	17,500	-56,5
4 Jul	yes	0,567	21,1	yes	15,683	-51,4
5 Jul	yes	0,417	21,2	yes	18,269	-58,1
6 Jul	yes	0,489	23,5	yes	16,938	-60,1
7 Jul	yes	0,264	26,3	yes	15,174	-53,3
8 Jul	yes	0,492	26,7	yes	11,703	-57,3
9 Jul	yes	0,205	21,4	yes	16,947	-57,5
10 Jul	no	0,272	24,3	yes	15,294	-57,1
11 Jul	yes	0,229	27,6	yes	15,485	-57,8

**General Site Information:
Meteorological, Geological
And Utilities Data**

Project No.: 01000
KPI Doc No.: 01000-EM-SPC-003
Contr. Doc No.:
Rev.: R00
Page: 11 / 21

12 Jul	yes	0,307	29,5	yes	16,880	-59,8
13 Jul	yes	0,417	24,8	yes	17,973	-61,9
14 Jul	yes	0,299	23,1	yes	15,177	-54,1
15 Jul	yes	0,291	21,9	yes	10,828	-42,3
16 Jul	yes	0,400	28,2	yes	15,829	-56,6
17 Jul	no			yes	10,414	-43,5
18 Jul	yes	0,286	29,0	yes	16,842	-61,5
19 Jul	yes	0,341	31,7	yes	16,218	-59,5
20 Jul	yes	0,168	29,0	yes	17,743	-60,8
21 Jul	yes	0,541	29,4	yes	13,992	-54,8
22 Jul	yes	0,361	27,6	yes	16,075	-56,0
23 Jul	yes	0,205	26,5	yes	16,114	-58,5
24 Jul	yes	0,184	28,6	yes	17,006	-59,3
25 Jul	yes	0,229	30,7	yes	16,151	-57,9
26 Jul	yes	0,277	29,0	yes	14,684	-58,4
27 Jul	yes	0,564	30,1	yes	16,616	-65,1
28 Jul	yes	0,191	33,8	yes	16,601	-62,6
29 Jul	yes	0,300	27,6	yes	16,472	-63,5
30 Jul	yes	0,343	23,3	yes	17,992	-67,2
31 Jul	yes	0,421	23,5	yes	17,246	-63,3

Temperature inversion in July 2009

Date	2009					
	Near the ground (0 to 1 km)			Atmospheric (1 to 30 km)		
	Inversion	Height (m)	Temperature (°C)	Inversion	Height (m)	Temperature (°C)
1 Jul	yes	0,552	25,1	yes	13,024	-48,6
2 Jul	yes	0,415	17,7	yes	13,055	-45,9
3 Jul	yes	0,245	20,9	yes	13,816	-50,0
4 Jul	yes	0,224	20,4	yes	14,044	-50,4
5 Jul	yes	1,000	14,1	yes	13,439	-51,3
6 Jul	yes	0,208	25,2	yes	12,729	-51,9
7 Jul	yes	0,500	20,1	yes	14,291	-49,4
8 Jul	yes	0,267	19,2	yes	12,790	-48,9
9 Jul	yes	0,170	22,5	yes	14,593	-54,4
10 Jul	yes	0,347	26,4	yes	13,714	-52,0
11 Jul	yes	0,201	24,9	yes	12,860	-51,7
12 Jul	yes	0,513	27,6	yes	12,281	-47,2
13 Jul	no			yes	12,236	-45,7
14 Jul	no			yes	15,361	-54,8
15 Jul	yes	0,329	28,0	yes	15,167	-54,0
16 Jul	yes	0,326	26,3	yes	23,074	-38,7
17 Jul	yes	0,238	30,2	yes	12,066	-50,8
18 Jul	yes	0,276	28,6	yes	12,653	-50,0
19 Jul	yes	0,391	29,0	yes	12,847	-49,1
20 Jul	yes	0,355	32,7	yes	13,186	-47,9
21 Jul	yes	0,266	31,9	yes	11,294	-42,1
22 Jul	yes	0,370	34,2	yes	23,019	-51,4
23 Jul	yes	0,304	26,8	yes	19,429	-56,7
24 Jul	yes	0,500	24,5	yes	12,137	-44,5
25 Jul	yes	0,401	26,1	yes	11,594	-43,4
26 Jul	yes	0,116	26,5	yes	14,513	-52,6

**General Site Information:
Meteorological, Geological
And Utilities Data**

Project No.: 01000
KPI Doc No.: 01000-EM-SPC-003
Contr. Doc No.:
Rev.: R00
Page: 12 / 21

27 Jul	yes	0,316	28,3	yes	12,479	-47,6
28 Jul	yes	0,289	26,2	yes	11,000	-41,2
29 Jul	yes	0,737	23,9	yes	13,811	-50,7
30 Jul	yes	0,443	25,5	yes	13,010	-50,8
31 Jul	yes	0,299	22,6	yes	13,115	-49,9

Temperature inversion for 5 Years

Date	2006		2007		2008		2009		2010	
Month	Near the ground (0 to 1 km)	Atmospheric (1 to 30 km)	Near the ground (0 to 1 km)	Atmospheric (1 to 30 km)	Near the ground (0 to 1 km)	Atmospheric (1 to 30 km)	Near the ground (0 to 1 km)	Atmospheric (1 to 30 km)	Near the ground (0 to 1 km)	Atmospheric (1 to 30 km)
Jan	100,0%	100,0%	74,2%	93,6%	96,8%	96,8%	87,1%	93,6%	96,8%	96,8%
Feb	100,0%	92,9%	92,9%	96,5%	100,0%	96,6%	92,9%	100,0%	85,7%	89,2%
Mar	90,4%	96,8%	96,8%	93,6%	100,0%	96,8%	93,6%	96,8%	77,4%	96,8%
Apr	93,4%	86,7%	96,7%	96,7%	73,4%	90,0%	90,0%	93,4%	76,7%	96,7%
May	96,8%	96,8%	96,8%	100,0%	80,7%	96,8%	83,9%	96,8%	71,0%	96,8%
Jun	86,7%	90,0%	93,4%	96,7%	90,0%	90,0%	96,7%	100,0%	96,7%	100,0%
Jul	90,4%	96,8%	96,8%	96,8%	96,8%	100,0%	93,6%	100,0%	93,6%	83,9%
Aug	96,8%	96,8%	96,8%	96,8%	100,0%	83,9%	83,9%	87,1%	96,8%	83,9%
Sep	83,4%	93,4%	90,0%	93,4%	63,4%	60,0%	93,4%	96,7%	90,0%	80,0%
Oct	93,6%	96,8%	100,0%	96,8%	100,0%	100,0%	93,6%	96,8%	83,9%	93,6%
Nov	90,0%	96,7%	90,0%	96,7%	100,0%	96,7%	76,7%	76,7%	83,4%	86,7%
Dec	51,7%	93,6%	90,4%	90,4%	96,8%	87,1%	77,5%	77,5%	96,8%	83,9%

Temperature inversion in avryst 2010

Date	2010					
	Near the ground (0 to 1 km)			Atmospheric (1 to 30 km)		
	Inversion	Height (m)	Temperature (°C)	Inversion	Height (m)	Temperature (°C)
1 Aug	yes	0,488	25,6	yes	12,593	-14,9
2 Aug	yes	0,342	32,8	yes	16,629	-35,0
3 Aug	yes	0,281	31,9	yes	15,101	-30,2
4 Aug	yes	0,154	33,3	yes	23,015	-32,3
5 Aug	yes	0,326	33,3	yes	12,767	-23,8
6 Aug	yes	0,189	33,4	no		
7 Aug	yes	0,428	31,2	yes	17,385	-31,2
8 Aug	yes	0,262	31,5	yes	12,245	-41,9
9 Aug	no			yes	12,173	-43,1
10 Aug	yes	0,358	33,3	no		
11 Aug	yes	0,560	32,1	yes	10,000	-10,6
12 Aug	yes	0,180	30,1	yes	12,785	-23,0
13 Aug	yes	0,509	30,3	yes	10,157	-31,7
14 Aug	yes	0,244	33,9	yes	17,000	-59,9

**General Site Information:
Meteorological, Geological
And Utilities Data**

Project No.: 01000
KPI Doc No.: 01000-EM-SPC-003
Contr. Doc No.:
Rev.: R00
Page: 13 / 21

15 Aug	yes	0,456	32,0	yes	16,802	-16,4
16 Aug	yes	0,768	23,5	no		
17 Aug	yes	0,586	26,4	yes	10,176	-11,9
18 Aug	yes	0,269	27,6	yes	11,289	-27,7
19 Aug	yes	0,229	30,0	yes	14,923	-29,9
20 Aug	yes	0,308	28,8	yes	12,834	-30,5
21 Aug	yes	0,348	19,6	no		
22 Aug	yes	0,316	12,7	yes	11,241	-12,9
23 Aug	yes	0,270	15,7	yes	13,991	-29,4
24 Aug	yes	0,206	19,7	yes	12,380	-49,4
25 Aug	yes	0,165	21,5	yes	12,681	-54,0
26 Aug	yes	0,194	23,8	yes	15,850	-47,9
27 Aug	yes	0,325	27,2	yes	14,266	-33,0
28 Aug	yes	0,183	23,3	yes	16,439	-44,9
29 Aug	yes	0,271	26,3	yes	10,351	-11,5
30 Aug	yes	0,431	29,4	no		
31 Aug	yes	0,229	32,2	yes	12,537	-34,4

1.10.3 Seismic design code shall be as follows: SNIP II-7-81. [TBD]

1.10.4 The region is subject to periodic inundation by storm surge from the Caspian Sea.

1.10.5 Maximum flood level Ural river – Atyrau

YEARS	Maximum level	
	m, (Baltic heights system)	Date
1981	-25,24	10.11.05
1982	-25,84	24.05
1983	-25,23	03.06
1984	-27,20	27.04
1985	-25,22	30.05
1986	-25,67	21.05
1987	-25,21	17.06
1988	-25,14	01.06
1989	-26,07	03.06
1990	-24,82	18.06
1991	-24,84	28.05
1992	-25,81	21.05
1993	-24,27	13.06
1994	-23,99	26.05
1995	-25,11	09.05
1996	-25,10	08.06
1997	-25,49	17.05
1998	-24,77	07.06
1999	-25,52	19.05
2000	-24,77	11.06
2001	-25,28	29.04
2002	-24,84	18.06
2003	-25,02	03.05
2004	-25,08	04.06
2005	-24,55	11.06
2006	-26,18	17.04

**General Site Information:
Meteorological, Geological
And Utilities Data**

Project No.: 01000
KPI Doc No.: 01000-EM-SPC-003
Contr. Doc No.:
Rev.: R00
Page: 14 / 21

2007	-24,87	22.06
2008	-25,77	14.05
2009	-26,16	21.04
2010	-25,96	12.05
2011	-25,13	23.05

1.10.6 Solar radiation: 881.5 W/m2

The temperature due to solar radiation to be considered is 80□

1.10.7 Climate area for construction : IV r (Abstracted from SNIP 2.04-01-2001)

1.10.8 Road climatic zone V (Abstracted from SNIP 2.01.07-85)

2 SITE GEOLOGICAL AND HYDROLOGICAL DATA

2.1 Site Locations IPC

The Karabatan Power Plant is a green field facility to be constructed in Karabatan, Atyrau Oblast (region) of Western Kazakhstan to supply electrical power to SEZ NIPT residents. The city of Atyrau, on the northern shore of the Caspian at the Ural River estuary, is the nearest industrial centre.

2.2 Existing Sites

The existing sites are open ground, part-covered by low scrub/weeds, with gently undulating surface levels varying between approximately -20m and -24m below Baltic Datum (-20m to -24m BD), which is approximately 2 - 6m above Caspian MSL.

Detailed topographic surveys of these areas, refer to the Section D of the ITB, Appendix 12 Site Preparation Plan.

2.3 Geotechnical Investigations At IPC Site IPC

The surface and subsurface soils of the site are salt-laden, of poor load-bearing quality, and may be susceptible to settlement and frost heave. Soil levelling works already done. Foundations for heavy equipment will almost certainly require piled support from a more competent soil layer 10m or so below ground level.

The Power Plant will operate in an area where flooding is probable. This flooding had caused with the time a deep salinification of the soil. That characteristic generating electrolyse phenomena and corrosion on buried elements, contractor shall have to prevent any damage on infrastructures like foundations, drainages, trenches, etc.

The upper layer of the soil is a sort of salty crust, the ground water is defined from -0.7m to -4.0m.

**General Site Information:
Meteorological, Geological
And Utilities Data**

Project No.:	01000
KPI Doc No.:	01000-EM-SPC-003
Contr. Doc No.:	
Rev.:	R00
Page:	15 / 21

During winter, the surface soils freeze down to a depth of up to 1.73m below the surface. This will be an important consideration affecting the burial depths of drains, utilities and concrete foundations.

Detailed geotechnical information will be provided.

2.4 Hydrological Studies

Due to strong winds causing an increase of the Caspian sea level, Power Plant will operate in an area where flooding is frequent. Contractor shall analyse site data and take all relevant mitigation measures to avoid any disturbance of operations if that phenomena occurs.

Hydrological studies of the risk of flooding of the Power Plant site from the River Ural and from Caspian seawater pileup shall be conducted by Contractor if required.

The roads shall be at + 3.0m above MSL and units at +2.0 to +2.5 m above MSL.

An optimisation of this solution can be proposed by the Contractor based on a hydrological study.

2.5 Earthquake

Earthquake Seismic Zone shall be 6 on the MSK-64 scale. Reference: Civil code SNiP RK 2.03-04-2001

3 SITE UTILITY SYSTEMS DATA

3.1 Utilities: General

It is anticipated that the following utilities systems will be required at the Power Plant site:

- Service water
- Pottable water
- Firewater
- Demineralised water
- Cooling water
- Plant Air
- Instrument Air
- Nitrogen
- Fuel Gas
- Electricity

All external utilities will be provided by third parties.

**General Site Information:
Meteorological, Geological
And Utilities Data**

Project No.: 01000
KPI Doc No.: 01000-EM-SPC-003
Contr. Doc No.:
Rev.: R00
Page: 16 / 21

3.2 Utilities: Water

3.2.1 Service Water

Service Water is to be clarified and filtered water from the raw water.

Service water delivery conditions at user's inlet battery limit at grade shall be:

CONDITIONS	Min	Normal	Max	Mech Design
Supply pressure, barg	6	6.5	7.3	11
Supply temperature, °C	5.6	20	30	80

3.2.2 Potable Water

Potable water hygienic requirements and quality control are shown in the table below:

Description	Potable Water Specification
Odor at 20°C, score, no more than	Undetectable
Turbidity under the standard scale, no more than NTU	< 1
Color, degrees, no more than	< 15
Lactose –positive Bacillus Coli, no more than	Not detectable
Coliphage, no more than	Not detectable
Helminth eggs(ascarids, whipworms, toxocars, oascils)	Not detectable
PH value	6.5~8.5
Suspended substances, mg/L no more than	< 1
Mineral content, mg/dm ³ , must not exceed solids including Chlorides Sulfate	250
Hardness total mg-equivalent/L, no more than	450(as CaCO ₃)
Chemical oxygen demand, mgO ₂ /dm ³ , range COD,	3
Aluminium, mg/L	0.2
Beryllium, mg/L	0.002
Boron, mg/L	0.5
Manganese, mg/L	0.1
Copper, mg/L	1.0
Molybdenum, mg/L	0.07
Arsenic, mg/L	0.01
Nickel, mg/L	0.02
Nitrates(NO ₃), mg/L	10
Lead, mg/L	0.01
Ferrum, including, Ferric chloride(Fe), mg/L	0.3
Mercury, mg/L	0.001
Selenium, mg/L	0.01
Phenol, mg/L	0.002
Residual chlorine, mg/L	0.3
Zink, mg/L	1.0
Sodium, mg/L	200
Chrome, mg/L (Cr ³⁺) (Cr ⁶⁺)	0.05
Cadmium, mg/L	0.005

**General Site Information:
Meteorological, Geological
And Utilities Data**

Project No.: 01000
KPI Doc No.: 01000-EM-SPC-003
Contr. Doc No.:
Rev.: R00
Page: 17 / 21

Barium, mg/L	0.7
Antimony, mg/L	0.005
Dichloromethane, mg/L	0.02
Dichlorvos, mg/L no more than	0.001

Potable Water delivery conditions at user's inlet battery limit at grade shall be:

CONDITIONS	Min	Normal	Max	Mech Design
Supply pressure barg	3.0	4.0	6.0	10.0
Supply temperature, °C	5.6	20	30.0	80.0

3.2.3 Fire Water

Fire water quality is same as make-up cooling water specification.

Firewater conditions at user's inlet battery limit at grade shall be:

CONDITIONS	Min	Normal	Max	Mech Design
Supply pressure barg,	8.0	10	12.0	16.0
Supply temperature, °C	5.6	20.0	30.0	80.0

3.2.4 Demineralised Water

Demineralised Water quality

Item	Units	Value
Total hardness, as CaCO ₃	ppm	< 0.05
Total dissolved solids	ppm	< 0.1
Electrical conductivity at 25 °C	µS/cm	< 0.2
Total CO ₂		Saturated
Total organic content	ppm	< 0.2
Oxygen	ppm	Saturated
Colour	Pt/Co	Colourless
Chlorides, as Cl ⁻	ppm	< 0.01
Fe	ppm	< 0.01

**General Site Information:
Meteorological, Geological
And Utilities Data**

Project No.: 01000
KPI Doc No.: 01000-EM-SPC-003
Contr. Doc No.:
Rev.: R00
Page: 18 / 21

SiO ₂	ppm	< 0.01
Oily matter	ppm	< 0.05
Na	ppm	< 0.01
Cu	ppm	< 0.003

Demineralised Water at user's inlet battery limit at grade

CONDITIONS	Pressure bar g			Temperature °C		
	min	nor	max	min	nor	max
Demineralized Water	6.0	6.5	7.3	10	20	60

3.2.5 Cooling Water

Cooling water quality are shown in the table below:

Compounds	Circulating Water (closed loop for process cooling) and Make - up water to cooling tower specification
NH ₄	<0,5
K	<10
Na	<10
Mg	<2
Ca	<5
Sr	<0,5
Ba	<0,5
CO ₃	<0,2
HCO ₃	<10
NO ₃	<40
Cl	<20
F	<0,2
SO ₄	<15
SiO ₂	<2
Boron	<2
TDS	<100
Ph	5-8
Fer ++	<0,1
Mn	<0,05
Al	<0,05

Cooling water delivery conditions at user's battery limit at grade shall be:

CONDITIONS	Min	Normal	Max	Mech Design
Supply pressure barg	5.0	5.4	7.0	10.5

**General Site Information:
Meteorological, Geological
And Utilities Data**

Project No.: 01000
KPI Doc No.: 01000-EM-SPC-003
Contr. Doc No.:
Rev.: R00
Page: 19 / 21

Return pressure barg	3.0	3.4	5.0	10.5
Supply temperature, °C			35	80
Return temperature, °C			45	80

3.3 Utilities: Air

Plant Air

Plant air delivery conditions

Item	Value
Design Dew point	Plant Air is not dried. Saturated moist air at Operating temperature and operating temperature
Dust	≤5 μm
Oil content	≤1 mg/m ³

3.5.1 Instrument and Plant Air

Instrument air delivery conditions at user's inlet battery limit at grade:

Item	Value
Dew point	-40 °C at 8 barg
Dust content	Class 2 (ISO 8573-1 2001)
Particulates	≤5 μm
Oil content	≤0.1mg/m ³

3.5.2 Utilities: Nitrogen

Nitrogen delivery conditions at user's inlet battery limit at grade:

CONDITIONS	Min	Normal	Max	Mech Design
Header pressure, barg	7.0	8	9	10
Supply temperature, °C	Amb	Amb	50	80

Nitrogen quality

Item	Value
Purity	≥ 99.99 % volume
Oxygen	< 5 ppm volume
Carbon monoxide	< 1 ppm volume
Carbon dioxide	< 1 ppm volume
Water	< 5 ppm volume
Particulates Dust- and oil	free
Dew point at atmospheric pressure	-65 °C

**General Site Information:
Meteorological, Geological
And Utilities Data**

Project No.: 01000
KPI Doc No.: 01000-EM-SPC-003
Contr. Doc No.:
Rev.: R00
Page: 20 / 21

3.4 Utilities: Fuels

3.6.1 Gas

Gas for consumption in Power Plant (Fuel Gas) will consist of Natural Gas taken from the NCC (Makat – North Caucasus) pipeline, augmented by offgases produced in the process units. The specification and estimated composition of the Natural Gas is as tabulated below.

No.	Characteristics	Requirements of Standard STRK 1666-2007	Actual
1	Heat of combustion, the lowest value, kcal/m ³ at not less than 20 °C, 101,325 kPa	7600	8105
2	Range of Wobbe number not more than, kcal/m ³	9850-13000	11705
3	Mass concentration of hydrogen sulphide not more than, g/m ³	0,007	0,0014
4	Mass concentration of mercaptansulphur, not more than, g/m ³	0,016	0,0021
5	Volume fraction of oxygen not more than, %	1	0,006
6	Mass of mechanical impurities in 1 m ³ , g not more than	0,001	Nil
7	Component composition	In volume fraction, %	In mole fraction, %
	CH ₄	94,609	94,573
	C ₂ H ₆	2,569	2,584
	C ₃ H ₈	0,473	0,480
	i-C ₄ H ₁₀	0,076	0,078
	n-C ₄ H ₁₀	0,101	0,105
	i-C ₅ H ₁₂	0,020	0,021
	n-C ₅ H ₁₂	0,022	0,023
	C ₆ H ₁₄	0,073	0,080
	Nitrogen	1,367	1,364
	Carbon dioxide	0,684	0,686
8	*Water due point, °C at t _{gas} = 37,2 °C and P _{gas} = 50,7 kgs/cm ²	-4	
9	Water due point, °C at P _{gas} = 40kgs/cm ²	-6,6	
10	Density, kg/m ³ , at 20 °C and 760 mm of mercury (bulk density)	0,710	
11	Density, kg/m ³ , at 20 °C and 760 mm of mercury (pycnometric specific gravity)	0,712	

*Specifies the actual T and P values of gas at the time of measurement of dew-point temperature. Analyses is carried out by "Akkol" chemical laboratory.

Natural Gas conditions at the inlet battery limit of Power Plant

Parameter	Operating	Design
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**General Site Information:
Meteorological, Geological
And Utilities Data**

Project No.: 01000
KPI Doc No.: 01000-EM-SPC-003
Contr. Doc No.:
Rev.: R00
Page: 21 / 21

	Min	Norm	Max	
Pressure (bar g)	38	52	75	79
Temperature (°C)	20	50	70	95

*To be confirmed during engineering study.

Nature gas from battery limit through LACT is measured and decompressed.

In case of non-availability of Natural gas, Power Plant gas turbines shall be designed for ethene fuel firing.

3.5 Utilities: Electricity

Electrical power for initial start to be provided by grid operator.

The voltage levels likely to be seen or utilized on site in Kazakhstan are as follows:

Grid Voltage Level: 110kV, 3Ø, 50Hz, 3 wires: (Earthing according to the Kazakhstan Grid)

Distribution, MV Voltage Level: 10kV, 3Ø, 50Hz, 3 wires: Resistance earthed Distribution,

LV Voltage Level: 690V, 3Ø, 50Hz, 3 wires: Direct to earth

Distribution, LV Voltage Level: 400/230V, 3Ø, 50Hz, 4 wires: Direct to earth Motor Applications:

Motors above 400kW 10 kV, 3Ø, 50Hz,

Motors from 50kW to 400kW 690 V, 3Ø, 50Hz,

Motors from 0kW to 50kW 400 V, 3Ø, 50Hz,

Elsewhere,

Lighting 230Vac, 1Ø, 50Hz, 2 wires

Small Power Circuits and sockets 230Vac, 1Ø, 50Hz, 2 wires +

earth Heat Tracing Circuits 230Vac, 1Ø,

50Hz, 2 wires Control Circuits – MV Motors 110Vdc

Control Circuits – LV Motors 230Vac, 1Ø,

50Hz Control Circuits – Instrumentation 230Vac, 1Ø, 50Hz

UPS Instrumentation Power 230Vac, 1Ø, 50Hz,

2 wires DC Power – Electrical 110Vdc

DC Power – Instrumentation 24Vdc